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Education

Ph.D. Physics, California Institute of Technology 2016
Thesis: [Naturalness confronts nature: Searches for supersymmetry with the CMS detector in pp collisions at \$\sqrt{s} = 8\$ and 13 TeV](#)
Advisor: Maria Spiropulu
M.S. Physics, California Institute of Technology 2015
S.B. Physics and Mathematics, Massachusetts Institute of Technology 2010
Thesis: [Exotic antineutrino oscillations \(\$\bar{\nu}_e \rightarrow \bar{\nu}_\mu\$ \) in Double Chooz](#)
Advisor: Janet Conrad

Professional Experience

Associate Professor of Physics at UC San Diego, La Jolla, CA 2023–Present
Assistant Professor of Physics at UC San Diego, La Jolla, CA 2019–2023
[Lederman Fellow](#) at Fermilab, Batavia, IL 2016–2019
Technical Instructor in [Junior Lab](#) at MIT, Cambridge, MA 2010–2011

Fellowships and Awards

- [Breakthrough Prize in Fundamental Physics](#) (awarded to co-authors of CERN LHC Run-2 publications at ATLAS, CMS, ALICE, and LHCb) (2025)
- American Physical Society [Henry Primakoff Award for Early-Career Particle Physics](#) (2024)
- UCSD [Inclusive Excellence Award](#) (2023)
- [Sloan Research Fellowship](#) (2023)
- Research Corporation For Science Advancement [Cottrell Scholar Award](#) (2023)
- UCSD Undergraduate Research Hub [Outstanding Mentor Award](#) (2021)
- [DOE Early Career Award](#) (2020)
- William A. Lee Chancellor's Endowed Junior Faculty Fellowship II (2019–Present)
- LHC Physics Center Distinguished Researcher (2019)
- Fermilab Lederman Fellowship (2016–2019)
- NSF Graduate Research Fellowship (2011–2014)
- [Gates Millenium Scholar](#) sponsored by Hispanic Scholarship Fund (2006–2014)

Grants and External Funding

- Key Personnel for DOE Award [UCSD Experimental and Theoretical Particle Physics](#) (2024–2028).
- Co-PI of [PREP: Advancing Research and Education in AI/ML for Science \(AREAS\)](#) (2024–2027).
- Lead PI of RCSA Cottrell Scholars Collaborative [Hidden Figures in Physics and Astronomy](#) (2023–2025).
- PI of Sloan Research Fellowship (2023–2025).

- PI of RCSA Cottrell Scholar Award [Building a Better Foundation: Teaching Physicists and Machines How to Learn from Data](#) (2023–2026).
- Key Personnel for DOE Award [U.S. CMS SPRINT—A Scholar Program for Research Internship](#) (2023–2026).
- Key Personnel for DOE Award [Western Advanced Training for Computational High-Energy Physics \(WATCHEP\)](#) (2022–2027).
- Key Personnel and Institute PI for [NSF HDR Institute for Accelerated AI Algorithms for Data Driven Discovery \(A3D3\)](#) (2021–2026).
- Key Personnel for DOE Award for [HEP Consortium for Advanced Training \(HEPCAT\)](#) (2021–2024).
- Co-PI for DOE Award for [Real-time Data Reduction Codesign at the Extreme Edge for Science](#) (2021–2024).
- PI of DOE Early Career Award for [Real-Time Artificial Intelligence for Particle Reconstruction and Higgs Physics](#) (2020–2025).
- Co-PI of DOE Award for [FAIR Framework for Physics-Inspired Artificial Intelligence in High Energy Physics](#) (2020–2023).
- Co-PI of NSF Award for [Exploring Neural Network Processors for AI in Science and Engineering](#) (2020–2021).
- Key Personnel for [Investigating Heterogeneous Computing at the Large Hadron Collider Phase-II](#) sub-award of Internet2 NSF Grant [Exploring Clouds for Acceleration of Science \(E-CAS\)](#) (2020–2021).
- DOE QuantISED Award [Quantum Machine Learning and Quantum Computation Frameworks for HEP \(QMLQCF\)](#) (2018–2020).
- Fermilab LDRD Award for [Graph Neural Networks for Accelerating Calorimetry and Event Reconstruction](#) (2019–2021).
- Fermilab LDRD Award for [Accelerator Control with Artificial Intelligence](#) (2019–2021).

Selected Research Experience

Higgs Boson Measurements & Combinations

- Co-convenor of CMS Higgs $b\bar{b}/c\bar{c}$ subgroup (2025–2027).
- Co-author of CMS search for triple Higgs boson production in the six bottom quark final state [16] and contributor to first HHH white paper [32].
- Author of CMS search for boosted Higgs bosons decaying to two W bosons merged in a single jet [3].
- Author of CMS searches for boosted Higgs boson pair production in the four bottom quark final state [1, 78], all-hadronic $b\bar{b}V\bar{V}$ final state [7, 38], and CMS statistical combination of Higgs boson pair searches [14, 72] using full Run 2 data (2020–Present).
- Lead author of search for a highly boosted Higgs boson decaying to a bottom quark-antiquark pair using full Run 2 data [111] and contributions to the charm quark-antiquark search [57] and dedicated vector boson fusion search [31] (2017–Present).
- Co-author of first search for a highly boosted Higgs boson decaying to a bottom quark-antiquark pair using 2016 data, published in *Phys. Rev. Lett.* [135]; Adapted analysis for interpretation for differential gluon fusion Higgs boson p_T measurement [127]; Combination of this result with other channels led to an observation of $H(b\bar{b})$ decay [129] and other measurements [126] (2016–2018).

- Level-1 and high-level (software) trigger development for Higgs decaying to bottom quark-antiquark pairs produced in association with a Z boson decaying to neutrinos [129] (2016–2017).

Exotic Long-lived Particle and Dijet Searches

- Supervision of students and postdoctoral researchers performing searches for long-lived particles [6, 26, 39, 48, 58, 86, 95] (2019–Present).
- Co-convener of CMS Exotica Jets+X subgroup (2018–2020).
- Co-leader of analysis group searching for exotic, light spin-1 and spin-0 particles decaying to quarks [123, 124, 128, 136] (2017–2019).
- Co-leader of dijet resonance search group, including data scouting, wide resonance, and b-tagged resonance searches [27, 115, 117, 130–132, 137] (2016–2018).

Novel Machine Learning Algorithms for Physics

- Particle Data Group author responsible for review on machine learning [9] (2024–Present).
- Transformers for highly granular large-radius jet classification [2, 42] (2023–Present).
- Symmetry-preserving attention networks for boosted and resolved multi-Higgs-boson event reconstruction [12] (2022–Present).
- Co-author of reviews and book chapters on machine learning for high energy physics [30, 33, 35, 69, 79, 82] (2020–Present).
- Symbolic regression to find parametric functions to model binned distributions in high energy physics [23] (2024–Present).
- Co-convener of CMS Machine Learning Group (2023–2025).
- Self-supervised learning strategies for foundation models in high energy physics [19, 29, 34] (2023–Present).
- Development of frameworks for sharing findable, accessible, interoperable, and reusable (FAIR) data and models in high energy physics [51, 54, 56, 87] (2020–Present).
- Development of anomaly detection algorithms for new physics searches [18, 36, 53, 59, 88–90, 94, 96, 113] (2019–Present).
- Development of graph neural networks [35, 82] and transformers for particle-flow reconstruction [5, 20, 25, 45, 63, 68, 84, 99, 104], including explainable AI techniques [66, 91], and charged particle tracking [55, 81, 97, 108] (2019–Present).
- Supervision of students developing generative adversarial networks and autoencoders for fast sparse data generation in high energy physics [52, 61, 75, 93, 103, 110] (2019–Present).
- Contributor to the Snowmass 2022 Community Planning Exercise, including white papers on machine learning for Higgs boson pair production [80], graph neural networks [83], fast machine learning [85], and data science and machine learning in physics education [73]; Co-convener of the CompF04 subgroup on AI Hardware [62] and contributor to EF01/EF02 Higgs Boson [70], CompF03 Machine Learning [71], and Muon Collider reports [46].
- Studies of quantum machine learning and quantum computation frameworks for high energy physics, including charged particle tracking [98] (2018–Present).
- Development of deep neural networks for identifying boosted Higgs bosons decaying to $b\bar{b}$ and $c\bar{c}$ for the CMS experiment and beyond [67, 114, 119, 121, 134] (2017–Present).

Fast Machine Learning Inference for Physics

- Supervision of students developing tools for fault-tolerant edge neural networks [37, 44] (2023–Present).

- Neural architecture codesign [10, 21, 64] and surrogate models for fast resource estimation [11] (2023–Present).
- Efficient and interpretable transformers for charged particle tracking [15, 43] and jet tagging [13, 17, 28] (2023–Present).
- Set- and graph-based neural networks for jet tagging on FPGAs in the level-1 trigger [40] (2022–Present).
- Supervision of postdoctoral researchers and students developing an ASIC-based encoder for data compression in the CMS HGCAL [101] and improving its training via a differentiable Earth mover’s distance [50] (2020–Present).
- Fast machine learning scientific benchmarks [74] (2022–Present).
- Development of Quantized ONNX (QONNX) framework for representing arbitrary-precision neural networks [77] (2022–Present).
- Real-time AI on FPGAs for accelerator control [100] (2018–2021).
- Development of `hls4ml` for scientific low-power machine learning devices [4, 8, 41, 49, 65, 76, 92, 101, 102, 106] (2020–Present).
- Development of *Services for Optimal Network Inference on Coprocessors (SONIC)* [22, 24, 47, 60, 105, 112, 120, 125] (2018–Present).
- Creation of `hls4ml` tool for creating low-latency FPGA-based firmware implementations of machine learning algorithms [79, 81, 88, 107–109, 116, 122, 133] (2017–Present).
- R&D, including firmware development and hardware demonstration, for the CMS Global Correlator Trigger for the Phase-2 upgrade of the Level-1 trigger [118] (2017–Present).

Selected Publications, Reviews, Reports, Book Chapters, and Conference Proceedings

Selected publications, reviews, reports, book chapters, and conference proceedings to which I made a substantial contribution are listed here.

- [1] CMS Collaboration, “Improved results on Higgs boson pair production in the 4b final state”, Submitted to *Phys. Rev. D*, 2026, [arXiv:2604.27044](#).
- [2] CMS Collaboration, “Particle transformers for identifying Lorentz-boosted Higgs bosons decaying to a pair of W bosons”, Submitted to *J. High Energy Phys.*, 2026, [arXiv:2604.09809](#).
- [3] CMS Collaboration, “Search for Higgs boson production at high transverse momentum in the WW decay channel in proton-proton collisions at $\sqrt{s} = 13$ TeV”, Submitted to *J. High Energy Phys.*, 2026, [arXiv:2603.22233](#).
- [4] J. Gonski et al., “Machine Learning on Heterogeneous, Edge, and Quantum Hardware for Particle Physics (ML-HEQUPP)”, Submitted to *Phys. Rev. X Intell.*, 2026, [arXiv:2602.22248](#).
- [5] CMS Collaboration, “Full event interpretation with machine-learning-based particle-flow reconstruction in the CMS detector”, Submitted to *Eur. Phys. J. C*, 2026, [arXiv:2601.17554](#).
- [6] CMS Collaboration, “Search for b hadron decays to long-lived particles in the CMS endcap muon detectors”, *Phys. Rev. D* **113**, 012009 (2026), doi:10.1103/71dn-snzn, [arXiv:2508.06363](#).
- [7] CMS Collaboration, “Search for heavy scalar resonances decaying to Lorentz-boosted Higgs and Higgs-like bosons in the $b\bar{b}4q$ final state at $\sqrt{s} = 13$ TeV”, Submitted to *J. High Energy Phys.*, 2026, [arXiv:2602.00273](#).

- [8] J.-F. Schulte et al., “hls4ml: A Flexible, Open-Source Platform for Deep Learning Acceleration on Reconfigurable Hardware”, Accepted by *ACM Trans. Reconfigurable Technol. Syst.*, 2025, doi:10.1145/3801979, arXiv:2512.01463.
- [9] J. Duarte and U. Seljak and K. Terao, “Machine Learning”, Ch. 41 in Particle Data Group et al., “Review of particle physics”, 2025, arXiv:2512.11133, <https://pdg.lbl.gov/2025/reviews/rpp2025-rev-machine-learning.pdf>.
- [10] J. Weitz et al., “Surrogate Neural Architecture Codesign Package (SNAC-Pack)”, in 8th Machine Learning and the Physical Sciences Workshop at the 39th Annual Conference on Neural Information Processing Systems (2025), arXiv:2512.15998.
- [11] B. Hawks et al., “wa-hls4ml: A Benchmark and Surrogate Models for hls4ml Resource and Latency Estimation”, Accepted by *ACM Trans. Reconfigurable Technol. Syst.*, 2025, doi:10.1145/3787490, arXiv:2511.05615.
- [12] H. Li et al., “Reconstruction of boosted and resolved multi-Higgs-boson events with symmetry-preserving attention networks”, *J. High Energy Phys.* **11**, 119 (2025), doi:10.1007/JHEP11(2025)119, arXiv:2412.03819.
- [13] T. Legge et al., “Why Is Attention Sparse In Particle Transformer?”, in 8th Machine Learning and the Physical Sciences Workshop at the 39th Conference on Neural Information Processing Systems (2025), arXiv:2512.00210, https://ml4physicalsciences.github.io/2025/files/NeurIPS_ML4PS_2025_238.pdf.
- [14] CMS Collaboration, “Combination of searches for nonresonant Higgs boson pair production in proton-proton collisions at $\sqrt{s} = 13$ TeV”, Submitted to *J. Phys. G*, 2025, arXiv:2510.07527.
- [15] S. Govil et al., “Locality-Sensitive Hashing-Based Efficient Point Transformer for Charged Particle Reconstruction”, in 8th Machine Learning and the Physical Sciences Workshop at the 39th Conference on Neural Information Processing Systems (2025), arXiv:2510.07594.
- [16] CMS Collaboration, *Search for nonresonant triple Higgs boson production in the six b-quark final state in proton-proton collisions at 13 TeV*, CMS Physics Analysis Summary CMS-PAS-HIG-24-012 (2025), <https://cds.cern.ch/record/2945361>.
- [17] A. Wang et al., *Spatially Aware Linear Transformer (SAL-T) for Particle Jet Tagging*, 2025, arXiv:2510.23641.
- [18] CMS Collaboration, *Anomaly detection with AXOL1TL at the CMS Level-1 Trigger in 2024 and 2025*, CMS Detector Performance Note CMS-DP-2025-061 (2025), <https://cds.cern.ch/record/2942560>.
- [19] Z. Hao et al., “RINO: Renormalization Group Invariance with No Labels”, in 8th Machine Learning and the Physical Sciences Workshop at the 39th Conference on Neural Information Processing Systems (2025), arXiv:2509.07486.
- [20] CMS Collaboration, *Machine-Learned Particle-Flow Reconstruction with Transformer Models in CMS*, CMS Detector Performance Note CMS-DP-2025-033 (2025), <https://cds.cern.ch/record/2937578>.
- [21] J. Weitz et al., “Neural architecture codesign for fast physics applications”, *Mach. Learn. Sci. Technol.* **6**, 035009 (2025), doi:10.1088/2632-2153/adede1, arXiv:2501.05515.

- [22] D. Kondratyev et al., “SuperSONIC: Cloud-Native Infrastructure for ML Inferencing”, in *Practice and Experience in Advanced Research Computing* (2025), doi:10.1145/3708035.3736049, arXiv:2506.20657.
- [23] H. F. Tsoi et al., “SymbolFit: Automatic Parametric Modeling with Symbolic Regression”, *Comput. Softw. Big Sci.* **9**, 12 (2025), doi:10.1007/s41781-025-00140-9, arXiv:2411.09851.
- [24] H. Zhao et al., “Track reconstruction as a service for collider physics”, *J. Instrum.* **20**, P06002 (2025), doi:10.1088/1748-0221/20/06/P06002, arXiv:2501.05520.
- [25] F. Mokhtar et al., “Fine-tuning machine-learned particle-flow reconstruction for new detector geometries in future colliders”, *Phys. Rev. D* **111**, 092015 (2025), doi:10.1103/PhysRevD.111.092015, arXiv:2503.00131.
- [26] CMS Collaboration, “Dark sector searches with the CMS experiment”, *Phys. Rep.* **1115**, 448 (2025), doi:10.1016/j.physrep.2024.09.013, arXiv:2405.13778.
- [27] CMS Collaboration, “Enriching the Physics Program of the CMS Experiment via Data Scouting and Data Parking”, *Phys. Rep.* **1115**, 678 (2025), doi:10.1016/j.physrep.2024.09.006, arXiv:2403.16134.
- [28] A. Wang et al., “Interpreting and Accelerating Transformers for Jet Tagging”, in *7th Machine Learning and the Physical Sciences Workshop at the 38th Conference on Neural Information Processing Systems* (2024), arXiv:2412.03673, https://ml4physicalsciences.github.io/2024/files/NeurIPS_ML4PS_2024_189.pdf.
- [29] S. Katel et al., “Learning Symmetry-Independent Jet Representations via Jet-Based Joint Embedding Predictive Architecture”, in *7th Machine Learning and the Physical Sciences Workshop at the 38th Conference on Neural Information Processing Systems* (2024), arXiv:2412.05333, https://ml4physicalsciences.github.io/2024/files/NeurIPS_ML4PS_2024_222.pdf.
- [30] J. Duarte et al., “Machine learning for analysis and instrumentation in high energy physics”, in *Instrumentation and Techniques in High Energy Physics*, edited by D. Lincoln (World Scientific, 2024), p. 125, doi:10.1142/9789819801107_0005.
- [31] CMS Collaboration, “Measurement of boosted Higgs bosons produced via vector boson fusion or gluon fusion in the $H \rightarrow b\bar{b}$ decay mode using LHC proton-proton collision data at $\sqrt{s} = 13$ TeV”, *J. High Energy Phys.* **12**, 035 (2024), doi:10.1007/JHEP12(2024)035, arXiv:2407.08012.
- [32] H. Abouabid et al., “HHH whitepaper”, *Eur. Phys. J. C* **84**, 1183 (2024), doi:10.1140/epjc/s10052-024-13376-3, arXiv:2407.03015.
- [33] J. M. Duarte, “Novel machine learning applications at the LHC”, in *42nd International Conference on High Energy Physics* (2024), doi:10.22323/1.476.0012, arXiv:2409.20413.
- [34] Z. Zhao et al., “Large-Scale Pretraining and Finetuning for Efficient Jet Classification in Particle Physics”, in *22nd International Workshop on Advanced Computing and Analysis Techniques in Physics Research* (2024), arXiv:2408.09343.
- [35] E. Chien et al., “Opportunities and challenges of graph neural networks in electrical engineering”, *Nat. Rev. Electr. Eng.* **1**, 529 (2024), doi:10.1038/s44287-024-00076-z.
- [36] CMS Collaboration, *2024 Data Collected with AXOL1TL Anomaly Detection at the CMS Level-1 Trigger*, CMS Detector Performance Note CMS-DP-2024-059 (2024), <https://cds.cern.ch/record/2904695>.

- [37] O. Weng et al., “FKeras: a sensitivity analysis tool for edge neural networks”, *ACM J. Auton. Transport. Syst.* **1** (2024), doi:10.1145/3665334.
- [38] CMS Collaboration, *Search for highly energetic double Higgs boson production in the two bottom quark and two vector boson all-hadronic final state*, CMS Physics Analysis Summary CMS-PAS-HIG-23-012 (2024), <https://cds.cern.ch/record/2904879>.
- [39] CMS Collaboration, “Search for long-lived heavy neutral leptons decaying in the CMS muon detectors in proton-proton collisions at $\sqrt{s} = 13$ TeV”, *Phys. Rev. D* **110**, 012004 (2024), doi:10.1103/PhysRevD.110.012004, arXiv:2402.18658.
- [40] P. Odagiu et al., “Ultrafast jet classification on FPGAs for HL-LHC”, *Mach. Learn.: Sci. Technol.* **5**, 035017 (2024), doi:10.1088/2632-2153/ad5f10, arXiv:2402.01876.
- [41] J. Campos et al., “End-to-end codesign of Hessian-aware quantized neural networks for FPGAs”, *ACM Trans. Reconfigurable Technol. Syst.* **17** (2024), doi:10.1145/3662000, arXiv:2304.06745.
- [42] C. Li et al., *Accelerating Resonance Searches via Signature-Oriented Pre-training*, 2024, arXiv:2405.12972.
- [43] S. Miao et al., “Locality-Sensitive Hashing-Based Efficient Point Transformer with Applications in High-Energy Physics”, in *41st International Conference on Machine Learning*, Vol. 235 (2024), p. 35546, arXiv:2402.12535, <https://proceedings.mlr.press/v235/miao24b.html>.
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- [49] O. Weng et al., “Tailor: Altering skip connections for resource-efficient inference”, *ACM Trans. Reconfigurable Technol. Syst.* (2024), doi:10.1145/3624990, arXiv:2301.07247.
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- [52] A. Li et al., “Induced Generative Adversarial Particle Transformers”, in *6th Machine Learning and the Physical Sciences Workshop at the 37th Conference on Neural Information Processing Systems* (2023), arXiv:2312.04757, https://ml4physicalsciences.github.io/2023/files/NeurIPS_ML4PS_2023_213.pdf.

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- [54] R. Kansal et al., “JetNet: A Python package for accessing open datasets and benchmarking machine learning methods in high energy physics”, *J. Open Source Softw.* **8**, 5789 (2023), [doi:10.21105/joss.05789](https://doi.org/10.21105/joss.05789).
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- [61] R. Kansal et al., “Evaluating generative models in high energy physics”, *Phys. Rev. D* **107**, 076017 (2023), [doi:10.1103/PhysRevD.107.076017](https://doi.org/10.1103/PhysRevD.107.076017), [arXiv:2211.10295](https://arxiv.org/abs/2211.10295).
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- [66] F. Mokhtar et al., “Do graph neural networks learn traditional jet substructure?”, in *5th Machine Learning and the Physical Sciences Workshop at the 36th Conference on Neural Information Processing Systems* (2022), [arXiv:2211.09912](https://arxiv.org/abs/2211.09912), https://ml4physicalsciences.github.io/2022/files/NeurIPS_ML4PS_2022_57.pdf.

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Teaching

Computational Physics

- Instructor for Physics 105A: Mathematical & Computational Physics I (Winter 2026).
- Instructor for Physics 141/241: Computational Physics I: Probabilistic Models and Simulations (Spring 2023, Winter 2022).
- Instructor for Physics 142/242: Computational Physics II: PDE and Matrix Models (Winter 2025, Winter 2024, Spring 2022).

Machine Learning & Data Science

- Creator of and instructor for [Physics 139/239: Machine Learning in Physics](#) (Fall 2025, Spring 2024, Winter 2023).
- Lecturer at [SLAC Summer Institute](#) (Summer 2023).
- Lecturer at [US ATLAS Machine Learning Training](#) (Summer 2023).
- [NSF IAIFI Ph.D. Summer School](#) Lecturer on “Representations, networks, and symmetries for learning from particle physics data” and “Model compression and fast machine learning in particle physics” (Summer 2022).
- Particle physics domain mentor for data science capstone [DSC 180AB](#) (Fall 2020, Winter 2021, Fall 2021, Winter 2022).
- Guest speaker for Purdue Physics 324: Big Data Science II (Spring 2021)
- Guest speaker for [MIT 8.S50: Computational Data Science in Physics](#) (Winter 2021).
- Creator of the [LHC Physics Center Machine Learning Tutorial](#) (2017–2020).

Computer Science & Engineering

- Guest speaker for [CSE 237C: Validation and Testing of Embedded Systems](#) (Fall 2020, Fall 2021, Fall 2022, Fall 2023).

Particle Physics

- Creator of a [HEPCAT Lab Module on ML/AI on FPGAs](#) (Summer 2023).
- Co-instructor for Physics 239: Modern Collider Physics (Spring 2023).
- UCSD instructor of record for [Physics 239: Statistics in Particle Physics at the LPC](#) [Primary instructor: Harrison Prosper, Florida State University] (Fall 2021).

Introductory Physics

- Lead instructor for [Physics 2C: Fluids, Waves, Thermodynamics, and Optics](#) for 300+ undergraduate students (Winter 2020, Winter 2021, Spring 2021, Spring 2026).
- Teaching assistant in statistical and quantum mechanics at Caltech (2011–2012).

Seminar Courses

- Organizer of [Physics 191: Undergraduate Seminar on Physics](#) (Fall 2020) and guest speaker (Fall 2019).

- Guest speaker for Physics 261: Seminar on Physics Research at UC San Diego (Winter 2020, Winter 2021).
- Guest speaker for Thurgood Marshall College 2: Transfer Year Experience. (Fall 2021).
- Facilitator for [Taking Research into Your Classroom](#) Workshop at Waubensee Community College (2017).

Physics Lab

- Technical Instructor in [MIT Junior Lab](#), teaching third-year undergraduate physics students and maintaining the experiments (2010–2011).

Outreach

Community Outreach

- Exhibitor for UC San Diego Physics and Duarte Lab at the [Barrio Logan Science & Art Expo](#) and [Southeast Science & Art Expo](#) (2022–2024).
- Presenter at “Career Exploration Event” at [SAY San Diego](#) Teen Leadership Connections Camp at Lincoln High School on Tuesday, July 18 (2023).
- Speaker at “Meet a US CMS Professor” for US CMS Internship Program on Wednesday, August 10 (2022).
- Panelist on “Careers in STEM Teaching and Research in Higher Education Panel” for UC San Diego Physical Sciences Division Student Success Center on Monday, November 15 (2021).
- Invited faculty speaker at UC San Diego Physics Graduate Student Diversity Initiative Grad Recruitment Fair on Saturday, October 23 (2021).
- Speaker on “Undergraduate and graduate research opportunities” at UC San Diego SACNAS Chapter’s Community College Workshop on Saturday, November 21 (2020).
- Invited faculty speaker for [Young Physicists Program](#) at UC San Diego (2020, 2022).
- Co-director of [Saturday Morning Physics](#) and lecturer on [Symmetry, Antimatter, and Supersymmetry](#) at Fermilab (2018–2019).
- On-site coordinator for [Saturday Morning Physics](#) at Fermilab (2016–2018).

Mentoring

- Mentor for [STEMULATE Community College Research Program](#) (2024).
- Mentor for [Cal-Bridge program](#) (2021–Present).
- Mentor for [APS National Mentoring Community](#) (2021–Present).
- Mentor for [ENLACE binational summer research program](#) (2021–2023).
- Mentor for [U.S. CMS Mentorship Program](#) (2020–2022).

Advocacy

- U.S. LHC Users Association Annual Trip to U.S. Congress (2017, 2021).

Diversity Programs

- Co-chair of local organizing committee for APS Conference on Undergraduate Women and Gender Minorities in Physics (CU*iP) at UC San Diego (2023–2025).
- UC San Diego Physics representative for the Mentoring Project (2024–Present).
- [PATHS Scholar Program](#) Faculty Advocate (2022–Present).
- Creator and coordinator of A3D3 Postbaccalaureate Fellowship Program (2021–Present).
- UC San Diego Physics Department Equity, Diversity, and Inclusion Committee member (2020–2023).

- Faculty advisor for UC San Diego SACNAS Chapter (2022–Present).
- Coordinator of A3D3 NSF Institute Equity & Career Committee (2022–Present).
- US CMS Collaboration Diversity, Equity, and Inclusion Committee (2022–2024).
- Faculty advisor for UC San Diego Physics Department Graduate Student Diversity Initiative (2021).
- Mentor in the [SIST](#) internship program at Fermilab (2018)
- Graduate student ambassador for the [Fermilab SHPE chapter](#) (2018).
- Member of the [TARGET](#) program committee at Fermilab (2017–2019).
- Residential Facilitator for [MIT Interphase EDGE program](#) (2010).

Selected Conference, Workshop, and Seminar Presentations

- [LLM meets Physics Panel Discussion](#). Physics Colloquium. March 5, 2026. UC San Diego, La Jolla, CA, USA.
- [Searching for new physics at the energy and intelligence frontiers](#). NSF CI Compass Virtual Workshop. AI Meets CI: Intelligent Infrastructure for Major & Midscale Facilities. January 12, 2026.
- [Progress on AI-based jet tagging](#). Machine Learning for Jets (ML4Jets) 2025. August 18, 2025. Caltech, Pasadena, CA, USA.
- [Machine Learning Topics in A3D3](#). A3D3 All-Hands Meeting. August 16, 2025. Caltech, Pasadena, CA, USA.
- [hls4ml and friends: Tools for ML in trigger and DAQ](#). ML4FE Workshop. May 19, 2025. University of Hawaii, Honolulu, HI, USA.
- [From collisions to discoveries with machine learning at the energy frontier](#). Physics Colloquium. January 30, 2025. Caltech, Pasadena, CA, USA.
- [Machine learning opportunities for the next generation of particle physics](#). 41st International Conference on Machine Learning. July 24, 2024. Vienna, Austria.
- [Novel ML technique applications](#). 42nd International Conference on High Energy Physics. July 23, 2024. Prague, Czech Republic.
- [Exploring Higgs bosons at high energies: From jets as graphs to fast machine learning on FPGAs](#). Fundamental Physics Directorate Seminar. May 21, 2024. SLAC, Menlo Park, CA, USA.
- [Exploring Higgs bosons at high energies: From jets as graphs to fast machine learning on FPGAs](#). APS April Meeting. Session F10: Narain, Primakoff and Sakurai Prize Session. April 4, 2024. Sacramento, CA, USA.
- [Exploring Higgs bosons at high energies: From jets as graphs to fast machine learning on FPGAs](#). Frederica Darema Lecture Series. April 1, 2024. UC Davis, Davis, CA, USA.
- [Machine learning at the edge of particle physics](#). IAIFI Summer Workshop. August 14, 2023. Northeastern University, Boston, MA, USA.
- [Machine learning summary: From concept to practice](#). 15th International Workshop on Boosted Object Phenomenology, Reconstruction, Measurements, and Searches at Colliders. August 4, 2023. Lawrence Berkeley National Laboratory, Berkeley, CA, USA.
- [Building a better foundation: Teaching physicists and machines how to learn from data](#). Cottrell Scholar Conference. July 19, 2023. Tucson, AZ, USA.
- [Machine learning for triggering](#). Aspen Winter Conference: Prospecting for New Physics through Flavor, Dark Matter, and Machine Learning. March 28, 2023. Aspen Center for Physics, Aspen, CO, USA.

- [Recent advances in machine learning for high energy physics](#). Dark Interactions: New Perspectives from Theory and Experiment. November 16, 2022.
- [Measuring Higgs bosons using artificial intelligence](#). Physics Department Colloquium. November 3, 2022. University of California San Diego, La Jolla, CA, USA.
- [Measuring Higgs bosons using artificial intelligence](#). The Human Side of Science Lecture Series. November 1, 2022. University of San Diego, San Diego, CA, USA.
- [Model compression and fast machine learning in particle physics](#). IAIFI Summer School. August 2, 2022. Tufts University, MA, USA.
- [Representations, networks, and symmetries for learning from particle physics data](#). IAIFI Summer School. August 1, 2022. Tufts University, Medford, MA, USA.
- [CompF3: ML for Data Analysis Summary](#). Community Summer Study Snowmass 2022. July 19, 2022. University of Washington, Seattle, WA, USA.
- [CompF4: AI-Hardware Summary](#). Community Summer Study Snowmass 2022. July 19, 2022. University of Washington, Seattle, WA, USA.
- [A3D3 Postbaccalaureate Fellowship Program](#). Community Summer Study Snowmass 2022. July 18, 2022. University of Washington, Seattle, WA, USA.
- [Enabling the Higgs self-coupling measurement with highly energetic Higgs pairs in CMS](#). Joint Experimental-Theoretical Physics Seminar (Wine & Cheese). June 3, 2022. Fermilab, Batavia, IL, USA.
- [Accelerated Graph Neural Network Inference](#). Mini-workshop on Graph Neural Networks for Tracking Colocated with Connecting the Dots 2022. June 3, 2022.
- [Fast Machine Learning for Science](#). ML Performance: Benchmarking Deep Learning Systems (MLPerf-Bench) Tutorial at the 28th IEEE International Symposium on High-Performance Computer Architectures (HPCA 2022). April 3, 2022.
- [AI at the Edge of Particle Physics](#). HEP Seminar. November 17, 2021. Columbia University, New York, NY, USA.
- [AI at the Edge of Particle Physics](#). AI Distinguished Lecture Series. August 12, 2021. Argonne National Laboratory.
- [AI-Hardware Codesign for Real-Time Science](#). Harnessing Data Science for Autonomous Computing Materials Symposium. May 27, 2021. Ohio State University.
- [The Edge of Particle Physics](#). Department of Physics and Astronomy Colloquium. May 6, 2021. Cal State LA.
- [Graph Neural Networks for High Energy Physics](#). Elementary Particle Physics Seminar. April 28, 2021. University of Minnesota.
- [hls4ml: An open-source codesign workflow to empower scientific low-power machine learning devices](#). tinyML Research Symposium. March 26, 2021.
- [Graph neural network tracking on FPGAs](#). IRIS-HEP Topical Meeting. October 21, 2020.
- [Real-time AI in particle physics](#). ECE Graduate Seminar. October 16, 2020. Carnegie Mellon University, Pittsburgh, PA, USA.
- [Recent highlights from CMS](#). 53rd Annual Fermilab Users Meeting. August 12, 2020. Fermilab, Batavia, IL, USA.
- [Deep learning for Higgs and new physics at the LHC](#). High Energy Physics Division Seminar. April 8, 2020. Argonne National Laboratory, Lemont, IL, USA.
- [Low-latency machine learning inference on FPGAs](#). 2nd Machine Learning and the Physical Sciences Workshop at NeurIPS 2019. December 14, 2019. Vancouver, Canada.
- [Deep learning for Higgs and new physics at the LHC](#). Department of Physics and Astronomy Colloquium. November 11, 2019. University of Kansas, Lawrence, KS, USA.

- [Deep learning for Higgs and new physics at the LHC](#). High Energy Experiment Seminar. October 10, 2019. Boston University, Boston, MA, USA.
- [Machine learning on FPGAs for low-latency and high-throughput inference](#). eScience 2019. September 24-27, 2019. San Diego, CA, USA.
- [Deep learning on FPGAs tutorial](#). 1st Real Time Analysis Workshop. July 15-26, 2019. Institute Pascal, Université Paris-Saclay, Saint Aubin, France.
- [Machine learning using CERN open data](#). LHCP 2019. May 20–25, 2019. Benemérita Universidad Autónoma de Puebla, Puebla, Mexico.
- [Dark sector searches in CMS](#). LHCP 2019. May 20-25, 2019. Benemérita Universidad Autónoma de Puebla, Puebla, Mexico.
- [FPGA-accelerated machine learning inference for particle physics](#). Connecting the Dots 2019. April 2, 2019. Valencia, Spain.
- [Unlocking the potential of LHC data: boosted Higgs and deep learning](#). Particle Physics Special Seminar. February 20, 2019. University of Chicago, Chicago, IL, USA.
- [Boosted Higgs couplings and dark mediators with deep learning in CMS](#). Joint Experimental-Theoretical Physics Seminar (Wine & Cheese). December 14, 2018. Fermilab, Batavia, IL, USA.
- [Heavy flavour identification for boosted resonances and large cone jets in CMS](#). Machine Learning for Jets (ML4Jets) 2018. November 14–16, 2018. Fermilab, Batavia, IL, USA.
- [Boosted Higgs, dark matter, and deep learning](#). High Energy Physics Seminar. October 3, 2018. University of Pittsburgh, Pittsburgh, PA, USA.
- [hls4ml: Deploying Deep Learning on FPGAs for L1 trigger and Data Acquisition](#). Topical Workshop on Electronics for Particle Physics (TWEPP) 2018. September 17-21, 2018. KU Leuven Campus Carolus, Antwerp, Belgium.
- [Searches for Dark Matter Mediators with the CMS Detector](#). Conference on the Intersections of Particle and Nuclear Physics (CIPANP) 2018. May 29 - June 3, 2018. Hyatt Regency Indian Wells Conference Center, Indian Wells, CA, USA.
- [Fast inference of deep neural networks in FPGAs for particle physics](#). Research Techniques Seminar. April 24, 2018. Fermilab, Batavia, IL, USA.
- [Fast reconstruction and data scouting](#). Connecting the Dots 2018. March 20-22, 2018. University of Washington, Seattle, WA, USA.
- [Boosted Higgs in CMS](#). Higgs Couplings 2017. November 6–10, 2017. Heidelberg University, Heidelberg, Germany.
- [Unlocking the potential of CMS data: boosted Higgs, low-mass dijet resonances, and data scouting](#). High Energy Physics Seminar. October 30, 2017. Caltech, Pasadena, CA, USA.
- [Search for low-mass dijet resonances](#). TeVPA 2017. August 7-11, 2017. Columbus, OH, USA.
- [Inclusive search for boosted SM Higgs bosons using H to bb decays with the CMS detector at 13 TeV](#). APS DPF 2017. July 31 - August 4, 2017. Fermilab, Batavia, IL, USA.
- [Inclusive Higgs boson search using \$H \rightarrow b\bar{b}\$ decays](#). Collider Cross Talk. July 20, 2017. CERN, Geneva, Switzerland.
- [Introduction to CMS open data for boosted object tagging with machine learning applications](#). Data Science at High Energy Physics (DS@HEP) 2017. May 8–12, 2017. Fermilab, Batavia, IL, USA.

Service and Committee Work

Thesis Committees

- Ph.D. Thesis Committee for Anna Tifrea (UC San Diego) (2026–Present).
- Ph.D. Thesis Committee for Dhruba Ghosh (Virginia Tech) (2026–Present).
- Ph.D. Thesis Committee for Aneesh Baburaj (UC San Diego) (2025).
- Ph.D. Thesis Committee (Chair) for Zihan Zhao (UC San Diego) (2024–Present).
- Ph.D. Thesis Committee for Wyatt Joyce (UC San Diego) (2024–Present).
- Ph.D. Thesis Committee (Co-Chair) for Luke Fairbanks (UC San Diego) (2024–Present).
- Ph.D. Thesis Committee (Chair) for Daniel Primosch (UC San Diego) (2024–Present).
- Ph.D. Thesis Committee for Michael Pokornik (UC San Diego) (2024–Present).
- Ph.D. Thesis Committee for Olivia Weng (UC San Diego) (2024–2026).
- Ph.D. Thesis Committee for Qingyuan Chen (UC San Diego) (2023–Present).
- M.S. Thesis Committee (Chair) for Steven Tsan (UC San Diego) (2023–2024).
- Ph.D. Thesis Committee for Robert Lee (UC San Diego) (2023–Present).
- Ph.D. Thesis Committee (Chair) for Anthony Aportela (UC San Diego) (2023–Present).
- Ph.D. Thesis Committee (Chair) for Farouk Mokhtar (UC San Diego) (2023–2026).
- Ph.D. Thesis Committee for Yifan Xiang (UC San Diego) (2023–2024).
- M.S. Thesis Committee for Kyle Yang (UC San Diego) (2022).
- Ph.D. Thesis Committee for Yueqi Zhao (UC San Diego) (2023–Present).
- Ph.D. Thesis Committee (Chair) for Raghav Kansal (UC San Diego) (2022–2024).
- M.S. Thesis Committee for Paul Wang (UC San Diego) (2021).
- Ph.D. Thesis Committee for Christian Aganze (UC San Diego) (2021–2023).

Other Service

- Five Year Ad Hoc Review Committee for Marisa Abrajano, Provost, Earl Warren College (2026).
- APS Division of Particles and Fields (DPF) Meeting Energy Frontier Track Convener (2026).
- School of Physical Sciences Data Science for the Physical Sciences Task Force (2025–Present).
- San Diego Faculty Association Executive Board Member (2025–2027).
- APS Henry Primakoff Award for Early-Career Particle Physics Selection Committee (2024–2025).
- Scientific program committee (Track 2: Data Analysis - Algorithms and Tools) for the 22nd International Workshop on Advanced Computing and Analysis Techniques in Physics Research (ACAT) (2024).
- Scientific organizing committee for [2nd, 3rd, 4th, 5th, 6th, and 7th Fast Machine Learning for Science Conferences and Accelerated Artificial Intelligence for Big-Data Experiments Conference](#) (2020–Present).
- Program committee for [Fast Machine Learning for Science Workshop at ICCAD](#) (2023).
- Reviewer for the Neural Information Processing Systems (NeurIPS) Conference (2023).
- Local organizing committee for [Multi-Boson Interactions Conference](#) at UC San Diego (2023).
- Organizer of [NSF HDR Postbaccalaureate Workshop](#) at UC San Diego (2023).

- Referee for *Physical Review Letters*, *Physical Review D*, *Physical Review Research*, *Journal of High Energy Physics*, *Physics Letters B*, *European Physics Journal C*, *Machine Learning: Science and Technology*, *Computing and Software for Big Science*, *Applied Optics*, *Nuclear Instruments and Methods in Physics Research Section A*, and *Science Bulletin* (2019–Present).
- US CMS Advisory Board (2022–2024).
- US CMS Collaboration Annual Meeting Planning Committee (2021–2023).
- US CMS Collaboration Diversity, Equity, and Inclusion Committee (2022–2023).
- Reviewer for the 2022 Datasets and Benchmarks Track, and 3rd and 4th Machine Learning for the Physical Sciences Workshops at the Neural Information Processing Systems (NeurIPS) Conference (2020–2022).
- External reviewer for Swiss Data Science Center (SDSC), French National Research Agency (ANR), US Department of Energy (DOE) Early Career Research Program, and European Science Foundation (ESF) (2019–2022).
- Faculty advisor for UC San Diego Physics Department Graduate Student Diversity Initiative (2021).
- Guest Associate Editor for *Efficient AI in Particle Physics and Astrophysics* Research Topic in *Frontiers in Big Data and AI* (2021–2022).
- UC San Diego Physics Department Equity, Diversity, and Inclusion Committee member (2020–2022).
- UC San Diego Physics Department Graduate Admissions Committee member (2019–2022).

Supervision and Mentorship

Postdoctoral Researchers

- Melissa Quinnan (UC San Diego). Schmidt AI in Science Postdoctoral Fellow. CMS level-1 trigger, Higgs boson searches (2022–Present).
- Daniel Diaz (UC San Diego). LPC Distinguished Researcher. CMS level-1 trigger, long-lived particle searches (2021–Present).

Doctoral Students

- Ellison Scheuller (UC San Diego). WATCHEP fellow. Anomaly detection for CMS level-1 trigger (2025–Present).
- Jason Weitz (UC San Diego). WATCHEP fellow. Neural architecture codesign (2024–Present).
- Daniel Primosch (UC San Diego). Higgs boson pair production searches in CMS (2023–Present).
- Hyeon Seo Yun (Purdue University). A3D3 Mentorship Program (2023–Present).
- Zihan Zhao (UC San Diego). Self-supervised learning for jet tagging (2022–Present).
- Russell Marroquin Solares (UC San Diego). WATCHEP fellow. CMS level-1 trigger long-lived particle tagger (2022–Present).
- Haoyang (Billy) Li (UC San Diego). Higgs boson jet assignment, FAIR AI models (2022–Present).
- Olivia Weng (UC San Diego). Optimization of AI algorithms for FPGAs (2021–Present).
- Jieun Yoo (UIC). U.S. CMS Mentorship Program (2021–2022).
- Anthony Aportela (UC San Diego). Sloan fellow, HEPCAT fellow. Graph-based autoencoders for anomaly detection; Search for long-lived particles (2021–Present).
- Daniel Guerrero (University of Florida). U.S. CMS Mentorship Program (2020–2021).
- Farouk Mokhtar (UC San Diego). HDSI fellow, IRIS-HEP fellow. Machine learned particle-flow reconstruction; Search for boosted $H \rightarrow WW$ (2020–Present).

- Raghav Kansal (UC San Diego). IRIS-HEP fellow, LPC AI fellow, LPC graduate scholar. Graph-based generative adversarial networks for particle physics simulation; Search for boosted $HH \rightarrow b\bar{b}WW$ (2019–2024).
- Martin Kwok (Brown). Boosted Higgs boson search (2018–2020).
- Michael Krohn (CU Boulder). Boosted Higgs boson search, coupling measurement, and trigger development (2017–2018).
- Sean-Jiun Wang (University of Florida). Development and monitoring of triggers for the Higgs boson produced in association with a Z boson decaying to neutrinos (2017–2018).
- Andrzej Novak (RWTH Aachen University). Development of deep neural networks for boosted Higgs identification in CMS (2017–2019).
- Jiajing Mao (Caltech). Data scouting trigger stream development (2016–2018).
- Giulia D’Imperio and Federico Preiato (Sapienza University of Rome). Dijet searches (2016).

Masters Students

- Vivekanand Sahu (UC San Diego). Efficient particle transformer (2023–Present).
- Priyansh Bhatnagar (UC San Diego). Efficient transformers for particle physics (2023–2024).
- Rounak Sen (UC San Diego). Discretized GANs for particle physics (2023–2024).
- Prashant Krishnan Vaidyanathan (UC San Diego). Self-supervised learning for particle physics (2023–2024).
- Steven Tsan (UC San Diego). Diffusion models for particle physics (2022–2024).
- Venkat Krishnamohan (UC San Diego). Graph GANs for particle physics (2022–2023).
- Selwyn Reis Gomes (UC San Diego). Xilinx Alveo coprocessor support in `hls4ml` (2022–2023).
- Nirmal Thomas (UC San Diego). Ragged batching for graph neural network inference as a service (2022–2023).

Postbaccalaureate Students

- Luke McDermott (UC San Diego). Neural architecture codesign (2023–2024).
- Michael Miranda. US CMS Intern. CMS level-1 long-lived particle triggers (2023).
- Andrew Skivington (UC San Diego). A3D3 Postbaccalaureate Fellow. Anomaly detection for CMS level-1 trigger (2022–2023).

Undergraduate Students

- Trevin Lee (UC San Diego). LLP identification, SONIC, and MoE particle transformers (2024–Present).
- Timothy Legge (UC San Diego). Particle transformers for jet tagging interpretability (2025–Present).
- Samantha Rodriguez (San Diego City College). STEMULATE Program. Boosted decision trees for Higgs boson pair searches (2024).
- Juan D. Guadalupe-Rosado (University of Puerto Rico). US CMS Intern. Large-radius jet tagging level-1 trigger in CMS (2024).
- Lauren Cadle (UC San Diego). Symmetry-preserving attention networks for semi-merged top quarks (2024–Present).
- Peera Serumaga (UC San Diego). Symmetry-preserving attention networks for semi-merged top quarks (2024–Present).
- Adolfo Partida (UC San Diego). Symmetry-preserving attention networks for semi-merged top quarks (2024–Present).

- Julian Jackson (UC San Diego). Symmetry-preserving attention networks for semi-merged top quarks (2024–Present).
- Chang Liu (UC San Diego). Long-lived particle searches in CMS (2024–Present).
- Shuyang Zhang (UC San Diego). Long-lived particle searches in CMS (2023–Present).
- Jason Weitz (UC San Diego). Neural architecture codesign (2023–2024).
- Dmitri Demler (UC San Diego). Neural architecture codesign (2023–2025).
- Emily Pan (UC San Diego). GNNS for identifying long-lived particles (2023–Present).
- Aditya Sriram (UC San Diego). GNNS for identifying long-lived particles (2023–Present).
- Kaitlyn White (UC San Diego). Symmetry-preserving attention networks for multi-Higgs-boson reconstruction (2023–Present).
- Darius Chao (UC San Diego). Symmetry-preserving attention networks for multi-Higgs-boson reconstruction (2023–Present).
- Rohan Sachdeva (UC San Diego). Anomaly detection for CMS level-1 trigger (2023–Present).
- Jet Yue (UC San Diego). ABCD neural network for background estimation in CMS. (2023–Present).
- Evelyn Lorenzo (UC San Diego). PATHS Program Scholar (2022–2023).
- Zhaoyu Zhang (UC San Diego). Graphs GANs for particle physics (2022–2024).
- Mengke Zhang (UC San Diego). Undergraduate Research Award. Machine-learned particle-flow for the Compact Linear Collider (2022–Present).
- Anni Li (UC San Diego). IRIS-HEP fellow. Conditional generation with graph networks (2022–2023).
- Ricardo Efraín Parra Payano (Universidad Nacional de Ingeniería, Peru). APS National Mentoring Community (2022).
- Parvat Sapkota (University of Texas at Arlington). APS National Mentoring Community (2021).
- Saloni Agrawal (UC San Diego). EXPAND program. JetNet (2022).
- Carlos Pareja (UC San Diego). EXPAND program. JetNet (2022–2024).
- Thomas Sievert (UC San Diego). FMP program. Quantum machine learning for high energy physics (2021–2023).
- Brian Sheldon (UC San Diego). FMP program. Boosted Higgs boson searches at the Future Circular Collider (hadron mode) (2021–Present).
- John Choi (UC San Diego). FMP program. Long-lived particle identification for CMS level-1 trigger (2021–2023).
- Ishaan Kavoori (UC San Diego). FMP program. FAIR4HEP cookiecutter FAIR AI template (2021–2023).
- Simon Poon (UC San Diego). FMP program. Machine-learned missing transverse momentum for CMS level-1 trigger (2021–Present).
- Sukanya Krishna (UC San Diego). IRIS-HEP fellow. Real-time anomaly detection for jets (2021–Present).
- Jason Liang (UC San Diego). tinyML with Brevitas and FINN. (2021–2023).
- Tai Nguyen (UC San Diego). Undergraduate Research Scholarship. tinyML with Brevitas and FINN. (2021).
- Han Hiller (University of Washington). UM-CERN-REU program. Machine-learned missing transverse momentum for CMS level-1 trigger (2021).
- Rohan Shenoy (UC San Diego). Undergraduate Research Award. Improved autoencoder training for HGCal ASIC for data compression (2021–2023).
- John Chen (UC San Diego). AEP program. Variable-sized-input generative graph networks (2021).

- Jevon Suharnoko (UC San Diego). Dream fellow. Transpilation of PYTORCH-based neural networks to FPGA firmware with `hls4ml` (2021–2022).
- Rushil Roy (UC San Diego). FMP program. `tinyML` with `hls4ml` (2021–Present).
- Zichun Hao (UC San Diego). FMP program, Undergraduate Research Award. Lorentz-equivariant generative neural networks and $H \rightarrow WW$ tagging (2021–2023).
- Abdelrahman Elabd (University of Pennsylvania). IRIS-HEP fellow. Implementation of graph neural networks on FPGAs and integration into `hls4ml`. (2021–2022).
- Haifeng Ding (UC San Diego). FMP program. Higgs pair production sensitivity at future colliders (Snowmass study). (2021).
- Steven Tsan (UC San Diego). TRELs program. Unsupervised or semi-supervised anomaly detection algorithms for high energy physics. (2020–2022).
- Vesal Razavimaleki (UC San Diego). IRIS-HEP fellow. Implementation of graph neural networks on FPGAs. (2019–2021).
- Eric Moreno (Caltech). SURF program. Development of interaction and graph neural networks for boosted jet tagging with CMS open data. (2018–2020).
- Sydney Jenknins (University of Chicago). Compression and firmware implementation of interaction and graph neural networks for charged particle tracking at the LHC (2018).
- Eric Scotti (Brown University). Development of deep neural networks for boosted Higgs identification in CMS (2017–2018).

Press

- [“From particle flow to learned reconstruction at CMS”](#), CERN EP Newsletter, March 25, 2026
- [“Machine learning to reveal more about LHC particle collisions”](#), CERN Homepage, February 18, 2026
- [“Teaching CMS to trace particles”](#), CMS Physics Briefing, February 12, 2026
- [“AI hunts for the next big thing in physics”](#) by Matthew Hutson, IEEE Spectrum, February 3, 2026
- [“Physics vocabulary: AI edition”](#) by Emily Ayshford, Kim Hickok, and Chris Patrick, Symmetry Magazine, April 18, 2024
- [“Inside the hunt for new physics at the world’s largest particle collider”](#) by Dan Garisto, MIT Technology Review, February 20, 2024
- [“From life experience to research experience”](#) by Sarah Charley, Symmetry Magazine, February 6, 2024
- [“LHC Physicists can’t save them all”](#) by Laura Dattaro, Symmetry Magazine, November 14, 2023
- [“SDSC, UC San Diego Physicist Receives 2024 Henry Primakoff Award”](#) by Cynthia Dillon, SDSC News, October 26, 2023
- [“Javier Duarte Recognized for Inclusive Excellence”](#) by Michelle Franklin, UC San Diego School of Physical Sciences News, August 8, 2023
- [“Will AI make MC the MVP of particle physics?”](#) by R. M. Davis, Symmetry Magazine, July 18, 2023
- [“Four Early Career Professors at UC San Diego Awarded Sloan Research Fellowships”](#) by Michelle Franklin, Daniel Kane, Katherine Connor, UC San Diego Today, March 1, 2023
- [“Two UC San Diego Faculty Named 2023 Cottrell Scholars”](#) by Michelle Franklin, UC San Diego Today, February 9, 2023
- [“San Diego Supercomputer Center, UC San Diego Join Federal Effort to Train Next-Gen Physics Workforce”](#) by Cynthia Dillon, UC San Diego Today, February 7, 2023

- [“Machine Learning Shaking Up Hard Sciences, Too”](#) by Dan Garisto, IEEE Spectrum, October 7, 2022
- [“How physicists are probing the Higgs boson 10 years after its discovery”](#) by Emily Conover, Science News Magazine, June 29, 2022
- [“Probing Higgs self-coupling with boosted Higgs pairs”](#) by Artur Apresyan and Si Xie, Fermilab News, June 9, 2022
- [“MLPerf Results Show Advances in Machine Learning Inference Performance and Efficiency”](#), MLCommons, June 4, 2022
- [“MLPerf Results Highlight Advances in Machine Learning Inference Performance and Efficiency”](#), Inside HPC, April 6, 2022
- [“Double trouble Higgs”](#) by Sarah Charley, Symmetry Magazine, April 26, 2022
- [“Graph neural networks boost di-Higgs search”](#), CERN Courier, March 11, 2022
- [“Physicists Apply FAIRness to Data Studies”](#) by Kimberly Mann Bruch, UC San Diego News, February 15, 2022
- [“SDSC Builds AI-Focused ‘Voyager’ Supercomputer”](#), Intel, November 9, 2021
- [“Muon detector probes long-lived particles”](#), CERN Courier, November 5, 2021
- [“Hunting anomalies with an AI trigger”](#), CERN Courier, August 31, 2021
- [“A new window into the shadow world: Exotic particle decays in the muon detectors”](#), CMS Physics Briefing, August 30, 2021
- [“Physics, Computation Experts Help Earn \\$15M to Advance AI, Data Analysis”](#) by Cynthia Dillon, UC San Diego News, September 28, 2021
- [“San Diego Supercomputer Center Teams Up with Habana to Power Voyager”](#) by Cynthia Dillon, UC San Diego News, April 9, 2021
- [“Live long and prosper: Searching for the long-lived relatives of the Higgs boson”](#), CMS Physics Briefing, August 16, 2021
- [“Long-lived particles gather interest”](#) by James Beacham and Albert De Roeck, CERN Courier, July 21, 2021
- [“National Science Foundation Awards SDSC \\$5 Million to Develop Innovative AI Resource”](#) by Jan Zverina, UC San Diego News, July 1, 2020
- [“Boosting into the unknown: The highest energy Higgs bosons”](#), CMS Physics Briefing, May 20, 2020
- [“UC San Diego Physicist Making a Mark”](#) by Cynthia Dillon, UC San Diego Division of Physical Sciences News, July 1, 2020
- [“Fermilab scientists help push AI to unprecedented speeds”](#) by Javier Duarte, Sergo Jindariani, Ben Kreis and Nhan Tran, Fermilab News, January 29, 2019

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